

Домашнее задание №2

1. Установка с помощью файла docker-compose.yml

 docker-compose.yml – Блокнот

Файл Правка Формат Вид Справка

```
services:
  clickhouse:
    image: clickhouse/clickhouse-server:latest
    container_name: clickhouse-server
    hostname: clickhouse-server
    restart: unless-stopped

    ports:
      - "${CLICKHOUSE_HTTP_PORT:-8123}:8123"
      - "${CLICKHOUSE_TCP_PORT:-9000}:9000"
      - "${CLICKHOUSE_INTERSERVER_PORT:-9009}:9009"

    environment:
      - CLICKHOUSE_DB=${CLICKHOUSE_DB:-default}
      - CLICKHOUSE_USER=${CLICKHOUSE_USER:-default}
      - CLICKHOUSE_PASSWORD=${CLICKHOUSE_PASSWORD}
      - CLICKHOUSE_LOG_LEVEL=information

    volumes:
      - clickhouse-data:/var/lib/clickhouse
      - clickhouse-logs:/var/log/clickhouse-server

    ulimits:
      nofile:
        soft: 262144
        hard: 262144

    networks:
      - clickhouse-network

volumes:
  clickhouse-data:
    driver: local
  clickhouse-logs:
    driver: local

networks:
  clickhouse-network:
    driver: bridge
```

Запуск командой docker-compose up -d

```
$ docker-compose up -d
Network clickhouse-docker_clickhouse-network Creating
Network clickhouse-docker_clickhouse-network Created
Volume "clickhouse-docker_clickhouse-data" Creating
Volume "clickhouse-docker_clickhouse-data" Created
Volume "clickhouse-docker_clickhouse-logs" Creating
Volume "clickhouse-docker_clickhouse-logs" Created
Container clickhouse-server Creating
Container clickhouse-server Created
Container clickhouse-server Starting
Container clickhouse-server Started
```

Работающий инстанс:

**clickhouse-server**
clickhouse/clickhouse
8123:8123 → 9000:...

```
2026-07-15 20:44:12 clickhouse-server | /entrypoint.sh: create new user 'default' instead 'default'
2026-07-15 20:44:12 clickhouse-server | Processing configuration file '/etc/clickhouse-server/config.xml'.
2026-07-15 20:44:12 clickhouse-server | Merging configuration file '/etc/clickhouse-server/config.d/docker_related_config.xml'.
2026-07-15 20:44:12 clickhouse-server | Logging trace to /var/log/clickhouse-server/clickhouse-server.log
2026-07-15 20:44:12 clickhouse-server | Logging errors to /var/log/clickhouse-server/clickhouse-server.err.log
2026-07-15 20:44:13 clickhouse-server |
2026-07-15 20:44:13 clickhouse-server | /entrypoint.sh: create database 'myanalytics'
2026-07-15 20:44:15 clickhouse-server | Processing configuration file '/etc/clickhouse-server/config.xml'.
2026-07-15 20:44:15 clickhouse-server | Merging configuration file '/etc/clickhouse-server/config.d/docker_related_config.xml'.
2026-07-15 20:44:15 clickhouse-server | Logging trace to /var/log/clickhouse-server/clickhouse-server.log
2026-07-15 20:44:15 clickhouse-server | Logging errors to /var/log/clickhouse-server/clickhouse-server.err.log
```

2. Тестовый датасет загружен

```
select count() from trips where payment_type = 1;
```

Результат 1

```
select count() from trips where payment_type = 1
```

123 count()
1 1 850 287

3. Делаем скриншоты работающего инстанса и тестового запроса

4. Изменение настроек:

 *optimized_config.xml – Блокнот

Файл Правка Формат Вид Справка

```
<?xml version="1.0"?>
<clickhouse>
  <!-- ===== ПАМЯТЬ ===== -->
  <max_server_memory_usage>4294967296</max_server_memory_usage> <!-- 4GB -->

  <!-- ===== ПОДКЛЮЧЕНИЯ ===== -->
  <max_concurrent_queries>100</max_concurrent_queries>
  <max_connections>4096</max_connections>

  <!-- ===== ФОНОВЫЕ ПОТОКИ ===== -->
  <background_pool_size>16</background_pool_size>
  <background_schedule_pool_size>128</background_schedule_pool_size>
  <background_distributed_schedule_pool_size>16</background_distributed_schedule_pool_size>

  <!-- ===== КЭШИ ===== -->
  <mark_cache_size>1073741824</mark_cache_size> <!-- 1GB -->
  <uncompressed_cache_size>0</uncompressed_cache_size> <!-- ОТКЛЮЧЕН -->

  <!-- ===== ЛОГИ ===== -->
  <query_log>
    <database>system</database>
    <table>query_log</table>
    <partition_by>toYYYYMM(event_date)</partition_by>
    <flush_interval_milliseconds>1000</flush_interval_milliseconds>
  </query_log>
</clickhouse>
```

5. Отчёт о приросте/изменении производительности системы на основе проведённых настроек в файле *performance_report_ru.md*